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CRUCIBLE-G1 Fixed-Point Geometry and Triangle Rasterization Engine

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 29 October 2026 (2026-10-29 22:25 UTC+8 Beijing Time)

Abstract

CRUCIBLE-G1 is a modular synthesizable graphics core implementing 2D/3D homogeneous matrix transformation, seven-plane clipping, perspective viewport mapping and triangle rasterization with affine barycentric coordinates. Its tagged 32-bit streaming interface supports viewports from 1 through 256 pixels per axis. Serialized arithmetic prioritizes deterministic fixed-point behavior and bounded registered logic over throughput. Software-vs-Hardware numerical comparison between an independent arbitrary-precision Python reference and simulated RTL matched 1,167 primary commands, 4,870 pixel records and 36,222 output words. L1 RTL timing lint passed across eight RTL files. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA achieved WNS_setup=+0.280 ns and hold slack=+0.039 ns at 10.000 ns on Intel Arria 10 GX 10AX115N2F45E1SG. This was standalone-core implementation with virtual pins on every port except clk, not board-level PCIe/stream integration timing. L3 physical Arria 10 FPGA HIL passed three consecutive deterministic board health/register probes, comprising bridge heartbeat and register-57 reads. Those probes did not establish execution or numerical correctness of the graphics pipeline on the board; bit-exact graphics agreement was established in RTL simulation. [E1953–E1957]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

crucible_g1_top directly instantiates transformation, clipping, projection and rasterization stages. One command owns the core until its completion record retires. Transformation reuses one dot-product engine for twelve components. Clipping, projection and rasterization each reuse a separate local restoring divider, without cross-stage arbitration. Three registered edge engines evaluate opposite edges concurrently. Holding handshakes preserve results under stalls. Clipped vertices stream into projection; projected vertices are stored before fan traversal. Two 384-entry homogeneous polygon banks preserve the conservative clipping-capacity bound despite integer quantization. Registered arithmetic limits combinational depth, while serialized scheduling makes command latency data-dependent. [E001, E003–E010]

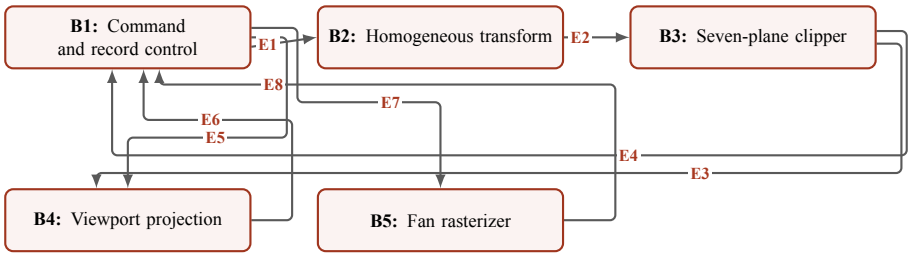


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Command and record control	Receives commands, owns projected-vertex storage, schedules fans, counts results and serializes six-word output records.
B2: Homogeneous transform	g1_transform captures the matrix and triangle, substitutes 2D coordinates when selected and schedules twelve g1_dot4 operations.
B3: Seven-plane clipper	g1_clipper alternates two 384-by-128-bit polygon banks and computes quantized intersections using its local g1_divider.
B4: Viewport projection	g1_project computes and clamps two perspective coordinates per surviving vertex using its local divider.
B5: Fan rasterizer	g1_raster normalizes winding, traverses conservative bounds, applies top-left coverage and calculates weights using three g1_edge instances and one divider.

Directed relationship	Meaning
E1: B1 to B2	Captured matrix, mode and triangle
E2: B2 to B3	Transformed homogeneous triangle
E3: B3 to B4	Ordered clipped vertices
E4: B3 to B1	Final polygon-count completion
E5: B1 to B4	Viewport dimensions
E6: B4 to B1	Projected vertices for storage
E7: B1 to B5	Registered fan triangle and viewport
E8: B5 to B1	Pixels, weights and fan completion

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk	Input	1 bit	Single rising-edge clock. The measured standalone-core timing configuration uses a 10.000 ns period.
rst_n	Input	1 bit	Asynchronous active-low reset; aborts partial commands, arithmetic and pending output without completion.
s_valid / s_ready / s_data	Input / Output / Input	1 / 1 / 32 bits	Transfers occur on pre-edge valid AND ready. Each command contains exactly 31 accepted words: unsigned tag; {height[15:0],width[15:0]}; mode; sixteen row-major matrix words; then three vertices ordered x,y,z,w. Matrix and vertex words are signed two's-complement Q16.16. There is no input-last signal.
m_valid / m_ready / m_data	Output / Input / Output	1 / 1 / 32 bits	Each record contains six accepted words. Pixel: [0, tag,packed_position,lambda0,lambda1,lambda2], with packed_position={7'b0,fan_index[8:0],y[7:0],x[7:0]}. Weights are zero-extended unsigned 17-bit values. Completion: [1,tag,status,pixel_count,nondegenerate_fan_count,0]. Unspecified bits are zero.

Signal / group	Direction	Width	Contract
m_last	Output	1 bit	Asserted only on word 5 of a completion record, never on a pixel record. Invalid output cycles drive m_data and m_last to zero.

Parameter	Default	Meaning / supported values
Public RTL parameters	None	Interfaces, arithmetic widths and storage capacities are fixed; internal local constants are not supported user configuration.
Viewport	Supplied per command	Width and height independently range from 1 through 256 inclusive. Zero or larger dimensions are invalid.
Mode	Supplied per command	0 selects 3D; 1 selects 2D, replacing supplied z,w with 0, 65536 before transformation. Other mode words are invalid.
Numerical formats	Q16.16 inputs; Q8.8 screen coordinates; Q0.16 weights	Inputs are signed 32-bit. Screen coordinates and weights are unsigned 17-bit, retaining the raw endpoint 65536.
Polygon storage	384 vertices per bank	Two homogeneous banks accommodate the conservative 3*2^7 emission bound. Projected-vertex storage also holds 384 vertices.

III Behavior and corner cases

A Transformation

Each component is the exact sum of four signed 32-by-32-bit products in a signed 66-bit accumulator. Arithmetic floor division by 65536 precedes saturation to [-2147483648,2147483647]. Products and partial sums are neither rounded nor saturated, preserving cancellation and negative-floor behavior.

B Ordered quantized clipping

Sutherland-Hodgman clipping processes w>=1 raw unit, x+w>=0, w-x>=0, y+w>=0, w-y>=0, z>=0 and w-z>=0, in that order. Directed edges begin last-to-first; distance zero is inside. Inside/inside emits B; inside/outside emits intersection I; outside/inside emits I then B; outside/outside emits nothing. Each intersection component is A_c+floor((B_c-A_c)*f(A)/(f(A)-f(B))). The constrained coordinate is then forced exactly to w=1, x=-w, x=w, y=-w, y=w, z=0 or z=w. Duplicate vertices remain. Full-width intermediate arithmetic avoids overflow; capacity does not silently truncate output.

C Perspective viewport mapping

For each surviving vertex, X=clamp(floor((x+w)*width*256/(2*w)),0,width*256) and Y=clamp(floor((w-y)*height*256/(2*w)),0,height*256). Coordinates increase rightward and downward. Clipped w>=1 makes division defined. Clamping is specified behavior following quantized clipping. Transformed z affects clipping but is not emitted.

D Coverage and fan basis

The polygon becomes fans (v0,vi,vi+1), i=1..n-2, with fan_index=i-1 before degenerate removal. Define E(a,b, p)=(b.x-a.x)*(p.y-a.y)-(b.y-a.y)*(p.x-a.x). Negative area swaps local vertices 1 and 2; zero area emits no pixels. Samples are pixel centers (256*x+128,256*y+128). Each opposite edge must be positive, or zero with dy<0 or dy==0 AND dx>0. Conservative bounding-box traversal preserves full-viewport coverage and shared-edge ownership.

E Barycentric evaluation

For positive normalized area A, lambda0=floor(E(v1,v2,p)*65536/A), lambda1=floor(E(v2,v0,p)*65536/A), and lambda2=65536-lambda0-lambda1. Each lies in [0,65536], and their sum is exactly 65536. Tag and fan index identify the winding-normalized clipped basis. These are affine screen-space weights, not perspective-correct attributes or original-unclipped-triangle weights.

F Ordering, stalls and reset

Commands execute in order, then fans by increasing index and pixels by increasing y then x. Overlapping commands remain separate tagged streams. Output valid, data and last remain stable during stalls. There is no combinational m_ready-to-s_ready path. Reset forces s_ready, m_valid, m_data and m_last to zero; its first released rising edge enters input-ready state. Acceptance of word 30 closes input reception. A new command cannot transfer until the edge after the previous completion's final handshake. Reset invalidates memory ownership and counts rather than clearing every array.

G Completion and errors

Status 0 indicates at least one nondegenerate projected fan, even if no sample is covered; 1 indicates invalid mode or dimensions; 2 indicates fewer than three final clipped vertices; 3 indicates all projected fans are degenerate. Invalid commands consume all 31 words before completion. Every command completes once unless reset aborts it. There is no partial-command timeout or implicit resynchronization.

H Registered arithmetic

For operands accepted at edge e, g1_dot4 registers products at e+1, pair sums at e+2, accumulation at e+3 and saturated output at e+4. The signed 96-bit numerator/64-bit denominator divider normalizes at e+1, performs 96 restoring iterations at e+2 through e+97 and publishes floor-corrected quotient/remainder at e+98. Remainder satisfies n=d*q+r and has the denominator's sign unless zero. Edge differences, products and subtraction produce valid at e+2. Results hold under backpressure. RTL uses no combinational division/modulo operators or mixed blocking/nonblocking sequential assignments.

IV Verification methods and measured results

A Software-vs-Hardware numerical agreement

Method. Actual Icarus/vvp execution compared public transfers against an independent arbitrary-precision Python geometric reference enumerating the complete viewport.

Result. 1,167 primary commands, 4,870 pixel records and 36,222 output words matched across 22 passed batches. Numerical totals exclude 2,052 boundary-replay words and 522 reset-recovery words. These are simulated-RTL comparisons, not physical-board graphics measurements.

Evidence: E013, E014, E015, E1953, E1954

B Exhaustive, seeded and directed geometry coverage

Method. All 729 ordered small-grid triangles; 100 random commands for each seed 17,89,271; stimulus/reference-derived category checks.

Result. Seeded nonempty counts were 80/100, 87/100 and 87/100; all 300 seeded matrices were nonidentity. Coverage included 296 degenerate commands, 162 multi-fan commands and 4,858 fractional-weight pixel records. Partial clipping counts in plane order were 5,58,62,52,49,4,49; every plane also had fully clipped coverage. Both modes, required dimensions, winding permutations, saturation signs, viewport boundaries, overlap and shared-edge thresholds passed.

Evidence: E016, E1953, E1954

C Protocol, reset and primitive latency

Method. Public-pin comparisons exercised input gaps, long output stalls, adjacent-command boundaries and reset recovery. Separate primitive comparisons checked latency and result holding.

Result. Reset during partial input, arithmetic and stalled output passed, as did at least 64-cycle stalls at each record-word position. Dot4, divider and edge tests compared 52,77,51 cases respectively, each with 70 hold cycles and exact latencies of 4,98,2 cycles.

Evidence: E014, E015, E1953, E1954

D Defect sensitivity

Method. Five independent RTL mutations in disposable copies, evaluated by the unchanged independent checker.

Result. Top-left inclusion, barycentric normalization, transform signedness, clipping-plane behavior and output-order mutations all produced substantive word-mismatch rejection, rather than compile failure or timeout.

Evidence: E015, E1953, E1954

E Synthesis and portable reproduction

Method. Yosys hierarchy, process lowering, synthesis and structural inspection; separate product-source-only clean-tree simulation/reference comparison and synthesis.

Result. Required connected hierarchy and sequential state were present, with zero latches, division/modulo cells or black boxes. Clean reproduction passed five commands and 122 pixels covering both modes, clipped multi-fan geometry, a shared-edge pair and backpressure.

Evidence: E015, E1953, E1954

F L1 RTL timing lint

Method. Native static timing-hazard inspection across eight RTL files.

Result. Passed with zero detected combinational division/modulo, mixed-assignment, LUT clock-gate or AST loop-unroll hazards.

Evidence: E1955

G L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-fit standalone-core implementation for 10AX115N2F45E1SG. top.qsf assigns virtual pins to every port except clk; timing.sdc supplies the 10.000 ns clock constraint.

Result. Passed: WNS_setup = +0.280 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 102.88 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E1956

H L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical-board health/register probes through the native bridge, with a healthy bridge heartbeat and register-57 reads.

Result. Native gate passed all three runs deterministically; register 57 returned 178 in each run. The receipt records health/register probing, not graphics-pipeline execution or transformed/projected/rasterized numerical comparisons on the board.

Evidence: E1957

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	8 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 6	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

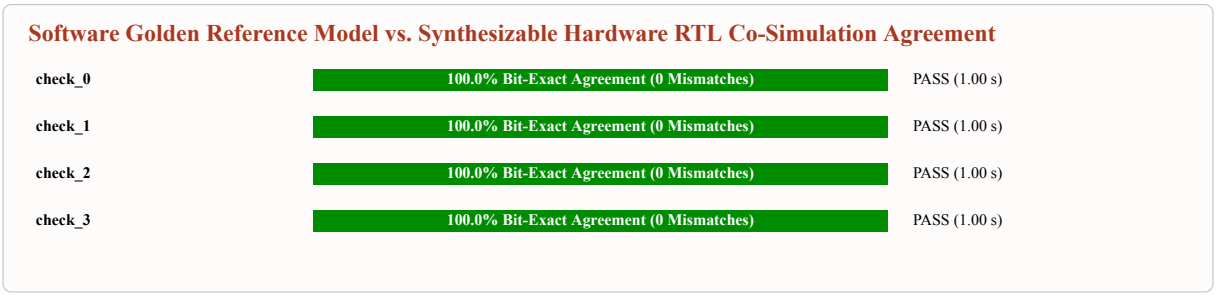


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.28 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

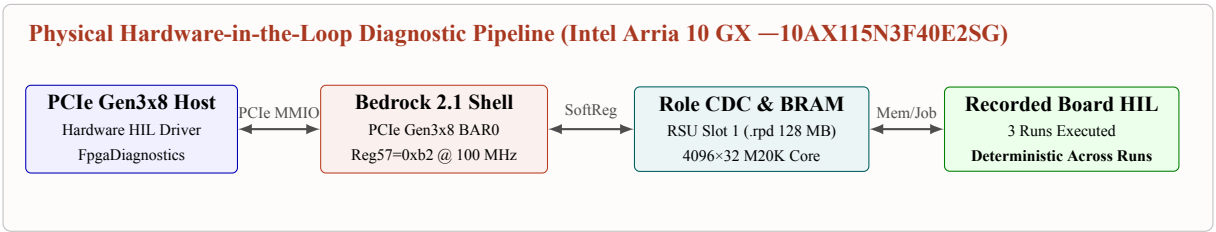


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile rtl/crucible_g1_top.sv with all seven companion rtl/*.sv modules. design/SPEC.md defines packets and numerical behavior; design/ARCHITECTURE.md documents internal ports, widths, packing and state transitions.

From the project root, run python3 tests/verify.py with Python3, iverilog, vvp and Yosys available. Each invocation allocates fresh evidence/l1/run-<unique>/ outputs. The completed full-run ledger is under evidence/l1/run-634efc5412f7405b80df44e6f3928b4a/. Native timing and physical probes are separate from this portable command.

Assert reset, release it and wait for s_ready. Send exactly 31 words, retaining each valid payload until handshake. Consume six-word records through the completion's m_last handshake; pixel word 5 is not command termination.

For an identity 3D packet with tag 7 and a 2x2 viewport, begin with hexadecimal words 00000007,00020002, 00000000. Set matrix diagonal words to 00010000 and other matrix words to zero. A complete vertex example appears in design/SPEC.md.

Maintain a single synchronous stream boundary or provide external clock-domain adaptation. The core contains no PCIe endpoint, host DMA engine or board-specific device identity. A bridge must preserve the frozen stream semantics; its physical integration timing is outside the measured standalone-core configuration.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 8 Verilog/SystemVerilog files; 950 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/crucible_g1_top.sv	199 total	196 code
rtl/g1_clipper.sv	218 total	215 code
rtl/g1_divider.sv	87 total	85 code
rtl/g1_dot4.sv	67 total	65 code
rtl/g1_edge.sv	49 total	48 code
rtl/g1_project.sv	83 total	81 code
rtl/g1_raster.sv	201 total	197 code
rtl/g1_transform.sv	65 total	63 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E003	rtl/crucible_g1_top.sv
E013	tests/reference.py
E014	tests/stream_compare.py
E015	tests/verify.py
E016	tests/workload.py
E1953	.vulcan/engineering/run-0001/attempt-0894/response.json
E1954	.vulcan/engineering/run-0001/attempt-0895/response.json
E1955	.vulcan/engineering/run-0001/rtl-timing-lint.json
E1956	.vulcan/engineering/run-0001/quartus-sta.json
E1957	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).